



Comparison Cheat-Sheet — "X vs Y"

Rapid revision · Every easy-to-confuse distinction in one place

Every "this vs that" interviewers love to test — scan this the night before.

1. 🗄️ Storage

S3 vs EBS vs EFS

	S3	EBS	EFS
Type	Object	Block	File (NFS)
Access	HTTP API, anywhere	One EC2 (one AZ)	Many EC2 across AZs
Scale	Unlimited	Up to 64 TiB/volume	Auto-scaling
Use	Backups, data lake, static sites	Boot/data disk for an instance	Shared filesystem

S3 storage classes (when to use)

Class	Use
Standard	Frequent access
Standard-IA / One Zone-IA	Infrequent, instant access (One Zone = 1 AZ, cheaper)
Glacier Instant / Flexible / Deep Archive	Archive: ms / minutes-hours / 12-48h retrieval
Intelligent-Tiering	Unknown/changing access patterns

2. 🖥️ Compute & Containers

EC2 pricing models

Model	Commitment	Discount	Use
On-Demand	None	—	Spiky / short-term
Reserved / Savings Plans	1–3 yr	up to ~72%	Steady-state
Spot	None (interruptible)	up to ~90%	Fault-tolerant, batch
Dedicated Host	Per host	—	Licensing / compliance

Container vs VM

	Container	VM
Isolates	A process (shares host kernel)	A full OS (own kernel)
Size / start	MBs / ms	GBs / minutes
Isolation	OS-level (namespaces+cgroups)	Hardware-level (stronger)

Docker vs Kubernetes — Docker builds/runs containers on **one host**; Kubernetes **orchestrates** many containers across **many hosts** (scheduling, scaling, self-healing).

Lambda vs EC2 — Lambda = serverless, event-driven, ≤15 min, pay per ms, auto-scale. EC2 = full VM you manage, always-on, any runtime/length.

3. 🗄️ Databases

DynamoDB vs RDS

	DynamoDB	RDS / Aurora
Model	NoSQL key-value/document	Relational (SQL)
Schema	Flexible	Fixed
Scale	Unlimited, automatic	Vertical + read replicas
Queries	By key/index (no joins)	Rich SQL, joins
Use	High scale, known patterns	Complex queries, transactions

RDS Multi-AZ vs Read Replica

	Multi-AZ	Read Replica
Purpose	High availability	Scale reads
Replication	Synchronous	Asynchronous
Readable?	✗ No	✓ Yes
Failover	Automatic, same endpoint	Manual promote

DynamoDB GSI vs LSI — GSI = different partition key, added anytime, own capacity, eventual reads. LSI = same partition key + alt sort key, created at table creation only, strong reads.

4. 🌐 Networking

Security Group vs NACL

	Security Group	NACL
Level	Instance / ENI	Subnet
State	Stateful	Stateless
Rules	Allow only	Allow + Deny (ordered)
Default	Deny in / allow out	Default allow all

ALB vs NLB vs GWLB

	ALB	NLB	GWLB
Layer	L7 (HTTP)	L4 (TCP/UDP)	L3
Routes on	Host/path/header	IP + port	IP packets
Use	Web apps, microservices	Ultra-fast, static IP	Virtual appliances

VPC Peering vs Transit Gateway — Peering = 1:1, non-transitive. TGW = hub-and-spoke, transitive, scales to many VPCs + on-prem.

Gateway vs Interface endpoint — Gateway = free, route-table target, **S3 & DynamoDB only**. Interface (PrivateLink) = ENI + private IP, most services, hourly + per-GB.

Public vs Private subnet — Public has a route `0.0.0.0/0 → IGW` ; private reaches internet only outbound via **NAT**.

5. 🚀 DNS & CDN

Route 53 Alias vs CNAME

	Alias	CNAME
Points to	AWS resources	Any DNS name
Zone apex?	✅ Yes	❌ No
Cost	Free (to AWS targets)	Charged

CloudFront vs S3 vs Global Accelerator

	CloudFront	S3	Global Accelerator
Role	CDN (cache at edge)	Object store	Route TCP/UDP to nearest endpoint
Caches?	✅ Yes	❌ No	❌ No
Layer	HTTP(S)	Storage	Network (static anycast IPs)

CloudFront Functions vs Lambda@Edge — Functions = lightweight JS, sub-ms, viewer-side (header/redirect). Lambda@Edge = full runtime, origin-side too, heavier logic.

6. 📧 Messaging & Events

SQS vs SNS vs EventBridge vs Kinesis

	SQS	SNS	EventBridge	Kinesis
Model	Queue (pull)	Pub/sub (push)	Event bus (routing)	Stream
Consumers	1 per message	Many subscribers	Rule targets	Many, replayable
Use	Decouple + buffer	Fan-out	Route by content/source	Real-time analytics, replay

SQS Standard vs FIFO — Standard = high throughput, at-least-once, best-effort order. FIFO = strict order + exactly-once, lower throughput (`.fifo`).

SQS vs Kinesis — SQS deletes a message once consumed (work queue). Kinesis is an ordered, **replayable** stream multiple consumers can re-read.

7. 🛡️ Security & Identity

IAM User vs Role

	User	Role
Credentials	Long-lived (keys/password)	Temporary (STS)
For	A person	Workloads, cross-account, federation
Preferred?	Avoid for apps	✅ Yes

SCP vs Permission Boundary — both only **cap** (never grant). SCP = org/OU/account-wide guardrail. Boundary = max for one user/role (safe delegation).

Identity-based vs Resource-based policy — identity-based attaches to a user/role; resource-based attaches to a resource and names a **Principal** (enables cross-account).

IAM evaluation: default deny → explicit **Deny wins** → must be allowed by $SCP \cap \text{boundary} \cap \text{identity/resource policy}$.

8. 📊 Monitoring & Audit

CloudWatch vs CloudTrail

	CloudWatch	CloudTrail
Answers	How is it performing ?	Who did what , when?
Data	Metrics, logs, alarms	API call audit log
Use	Monitoring, alerting	Auditing, compliance, forensics

EC2 status checks: System check = AWS infra (fix: stop/start). Instance check = the OS/instance (fix: reboot/fix OS).

9. 🔄 Git & GitHub

Merge vs Rebase

	Merge	Rebase
History	Preserved (branchy)	Rewritten (linear)
Merge commit	Yes	No
Safe on shared branches?	✅ Yes	❌ No

git fetch vs pull — fetch downloads without merging; pull = fetch + merge (or `--rebase`).

Fork vs Clone vs Branch — fork = your server-side copy; clone = a local copy; branch = a line of work in a repo.

10. 🧠 Memory Hooks

One-liners worth memorizing verbatim:

- **CloudWatch** = performance/health · **CloudTrail** = audit/who-did-what.
 - **Rebase private history, merge public history** ("rebase to clean up, merge to combine").
 - **Multi-AZ** = availability (sync, not readable) · **Read Replica** = scaling (async, readable).
 - **Alias** = AWS targets + works at apex + free · **CNAME** = any name, not at apex.
 - **SG** = stateful/instance/allow-only · **NACL** = stateless/subnet/allow+deny.
 - **SQS** = queue (one consumer) · **SNS** = pub/sub (many subscribers) · fan-out = SNS → many SQS.
 - **Explicit Deny** always wins in IAM; boundaries & SCPs **cap, never grant**.
 - **CloudFront** caches; **Global Accelerator** routes (no cache).
 - **Container** shares the host kernel; a **VM** ships its own.
 - **Spot** = cheapest + interruptible (2-min notice); **Reserved/Savings** = steady; **On-Demand** = flexible.
-

11. 🎯 Quick Decision Prompts

"Which service?" gut-checks:

- **Need a static IP on a load balancer?** → NLB (or Global Accelerator).
 - **Route by URL path / host?** → ALB.
 - **Decouple two services + buffer spikes?** → SQS.
 - **Notify many subscribers at once?** → SNS (fan-out to SQS).
 - **Key-value at massive scale, single-digit ms?** → DynamoDB.
 - **Complex joins / transactions / reporting?** → RDS/Aurora.
 - **Cache content near users / front S3 with HTTPS?** → CloudFront.
 - **Run code on an event, no servers, ≤15 min?** → Lambda.
 - **Give an EC2 app AWS permissions?** → IAM role (never hard-coded keys).
 - **Private access to S3 from a VPC?** → Gateway VPC endpoint (free).
 - **Highly available web tier?** → ALB + Auto Scaling across ≥2 AZs.
 - **Audit who deleted a resource?** → CloudTrail. **Why is CPU high?** → CloudWatch.
-

 Notes compiled as a quick-revision comparison cheat-sheet.